2.11 Einstein's Number Puzzle (Lab 2.1)

LAB ACTIVITY | 2.11.1: Einstein's Number Puzzle (Lab 2.1) | Full screen | 0 / 37

This number puzzle is usually attributed to Albert Einstein. Apparently he would write a number on a piece of paper, give it to someone else, and then let that person (or anyone else) pick any 3 digit number where the first and last digits differed by at least 2. After some math, regardless of which number was chosen, the result was always the number written down by Einstein before any number was chosen.

First, download this template and rename it `einstein.py`:

- template.py

Then, implement the following algorithm:

> **Set** `number` = a three-digit number input from the user. (You may assume that the user provides a three-digit number and that the first and last digits differ by at least 2.)
> **Set** `digit1` to the 100's digit of `number`.
> **Set** `digit2` to the 10's digit of `number`.
> **Set** `digit3` to the 1's digit of `number`.
> **Set** `rev_number` to the integer equal to these three digits in reverse order.
> **Set** `difference` to the positive difference between `number` and `rev_number`. (We want this result to always be positive, so use the `abs` function. e.g. `abs(4 - 8)` is 4.)
> **Set** `rev_diff` to the reverse of `difference`. (Note that this requires you to use steps 2-5 again but with `difference`.)
> **Print** the sum of `difference` and `rev_diff`.

For example, if the input is:

```
Number: 123↵
```

Then the output should (always) be:

```
1089
```

Notes:

- You may assume that the user of the program chooses an appropriate number.
- Please use only the operations we have studied in class so far (assignment, `%`, `//`, etc.).
- The first four storing steps (steps 2–5) in the algorithm compute and store the reverse of the number. For example, if the user enters 123, it will compute 321. As a challenge, you can try to do those four steps all in one statement.

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einstein.py

```
1   """CS 108 - Lab/Homework X.X
2
3   Describe the module here. Fix the lab number above and the name/date below.
4
5   @author: your name(s) here...
6   @date: semester, year
7   """
8
9   # Put your solution code here.
10
```

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```
History of your effort will appear here once you begin
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